

Technical Publication

Diesel Engine
16V 4000 G20
16V 4000 G40
16V 4000 G60
16V 4000 G80

Operating instructions
M015345/20E



Printed in Germany

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Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen.

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Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

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Imprimé en Allemagne

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Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

Impreso en Alemania

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El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

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Stampato in Germania

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

Con riserva di modifiche.

! **Safety Instructions**

Summary

A **Product Summary**

Engine layout, technical data

C **Operating Instructions**

Task descriptions

D **Operating Instructions**

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E **Troubleshooting**

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G **Task Description**

Inspection and maintenance tasks

K **Preservation**

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? **Keyword Index**

Alphabetical index of key terms

Wichtig / Important

Important / Importante / Importante

Bitte die Karte „Inbetriebnahmemeldung“ abtrennen und ausgefüllt an MTU Friedrichshafen zurücksenden.

Die Informationen der Inbetriebnahmemeldung sind Grundlage für den vertraglich vereinbarten Logistik-Support (Gewährleistung, Ersatzteile etc.).

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).

Veillez séparer la carte “Signalisation de mise en service” et la renvoyer à la MTU Friedrichshafen.

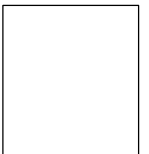
Les informations concernant la signalisation de mise en service constituent la base pour l'assistance en exploitation contractuelle (garantie, rechanges, etc.).

Rogamos separar la tarjeta “Aviso de puesta en servicio” y la devuelvan rellena a MTU Friedrichshafen.

Las informaciones respecto al aviso de puesta en servicio constituyen la base para el soporte logístico contractual (garantía, piezas de repuesto, etc.).

Staccare “Avviso di messa in servizio” e rispedirlo debitamente compilato alla MTU Friedrichshafen.

Le informazioni ivi registrate sono la base per il supporto logistico contrattuale (garanzia, ricambi, ecc.).



Postkarte
MTU Motoren- und Turbinen-Union
Friedrichshafen GmbH
Abteilung **SCSD**
88040 Friedrichshafen
Germany



Bitte in Blockschrift ausfüllen!
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¡A rellenar en letras de imprenta!
Scrivere in stampatello!



Motornr.: Engine No.: N° du monteur: N° de motor: Motore N::	Auftragsnr.: MTU works order No.: N° de commande: N° de pedido: N. commessa:
Motortyp: Engine model: Type du moteur: Tipo de motor: Motore tipo:	Inbetriebnahmedatum: Date put into operation: Mise en service le: Fecha de puesta en servicio: Messa in servizio il:
Eingebaut in: Installation site: Lieu de montage: Lugar de montaje: Installato:	Schiffstyp / Schiffshersteller: Vessel/type/class / Shipyard: Type du bateau / Constructeur: Tipo de bugue / Constructor: Tipo di Barca / Costruttore:
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Bemerkung: Remarks: Remarques: Observaciones: Commento:	

Inbetriebnahmemeldung

Commissioning Note

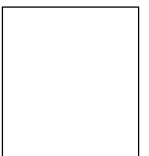
Notice de mise en service

Aviso de puesta en servicio

Avviso di messa in servizio

Amendment Service !

Any changes in contents will be sent to you in the form of an amendment,
provided you complete the reverse of this receipt card and return it to us.



Postcard

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Friedrichshafen GmbH
Department **VLT**

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Germany



Please mark all filed amendments here.



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Safety, Accident Prevention and Environmental Protection Instructions

These instructions must be read and followed by every person involved in operation, maintenance or transportation of the machinery plant.

General

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed.

This engine is a state-of-the-art product and conforms with all applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Non-compliance with the Safety Instructions

Correct Use

The engine is intended exclusively for the application specified in the contract or upon delivery. Any other use is considered improper use. The manufacturer will accept no liability for any resultant damage. The responsibility is borne by the user alone.

Correct use also includes observation of and compliance with the maintenance specifications.

Personnel Requirements

Work on the engine must only be carried out by properly qualified personnel.

The specified legal minimum age must be respected.

Responsibilities of the operating, maintenance and repair personnel must be specified.

Modifications or Conversions

Modifications made by the customer to the engine may affect safety.

MTU will accept no liability or warranty claims for any damage caused by unauthorised modifications or conversions.

Organisational Measures

This publication must be issued to all personnel involved in operation, maintenance, repair or transportation.

It must be kept at hand near the engine and accessible to all personnel involved in operation, maintenance, repair or transportation.

The personnel must be instructed on engine operation and repair by means of this publication, and in particular the safety instructions must be explained.

This is especially important for personnel who work on the engine only on an occasional basis. Such personnel must be given instructions repeatedly.

Spare Parts

Only genuine MTU spare parts must be used to replace components or assemblies. In event of any damage caused by the use of other spare parts, no liability or warranty claims vis-à-vis the engine manufacturer will be accepted.

Working Clothes and Protective Equipment

Wear proper work clothing for all work.

Depending on the kind of work, use additional protective equipment, e.g. protective goggles, protective gloves, protective helmet, apron.

Work clothing must be tight-fitting so that it does not catch on rotating or projecting components.

Do not wear jewellery (e.g. rings, chains, etc.).

Transportation

Lift the engine only with the lifting eyes provided.

Use only the transport and lifting equipment approved by MTU.

Take note of the engine centre of gravity.

The engine must only be transported in installation position.

In the case of special packaging with aluminium foil, suspend the engine on the lifting eyes of the bearing pedestal or transport with equipment suitable for heavy loads (forklift truck).

Prior to transporting the engine, it is imperative to install transportation locking devices for crankshaft and engine mounts.

Secure the engine against tilting during transportation. The engine must be especially secured against slipping or tilting when going up inclines and ramps.

Setting the Engine down after Transportation

Place the engine only on an even, firm surface.

Ensure appropriate consistency and load-bearing capacity of the ground or support surface.

Never place an engine on the oil pan unless expressly authorised by MTU on a case-by-case basis to do so.

Working with Laser Equipment

When working with laser equipment, always wear special laser-protection goggles.

Laser equipment can generate extremely intensive, concentrated radiation by the effect of stimulated emission in the range of visible light or in the infrared or ultraviolet spectral range. The photochemical, thermal and optomechanical effects of the laser can cause damage. The main danger is irreparable damage to the eyes.

Laser equipment must be fitted with the protective devices necessary for safe operation according to type and application.

For conducting light-beam procedures and measurement work, only the following laser devices must be used:

- Laser devices of classes 1, 2 or 3A,
- Laser devices of class 3B, which have maximum output only in the visible wavelength range (400 to 700 Nm), a maximum output of 5 mW, and in which the beam axis and surface are designed to prevent any risk to the eyes.

Engine Operation

When the engine is running, always wear ear protectors.

Ensure that the engine room is well ventilated.

Mop up any leaked or spilt fluids and lubricants immediately or soak up with a suitable bonding agent.

Exhaust gases from combustion engines are poisonous. Inhalation of poisonous exhaust gases is a health hazard. The exhaust pipework must be leak-free and must discharge the gases to atmosphere.

During engine operation, do not touch battery terminals, generator terminals or cables.

Inadequate protection of electrical components can lead to electric shocks and serious injuries.

When the engine is running, never release coolant, oil, fuel, compressed air or hydraulic lines.